

HHU-SyLo at SemEval-2026 Task 11: Logic in the Loop – Hybridizing LLMs and Theorem Provers for Robust Formal Reasoning

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Abstract

We present our system for SemEval-2026 Task 11 on reasoning disentanglement, separating syllogistic validity from semantic plausibility. We compare direct neural inference against two neuro-symbolic pipelines: translation to first-order logic and to syllogistic triples. By offloading inference to symbolic theorem provers, these hybrid models effectively mitigate content bias and improve logical fidelity.

1 Introduction

SemEval-2026 Task 11 (Valentino et al.) evaluates syllogistic reasoning by disentangling logical validity from semantic plausibility. LLMs frequently succumb to the content effect, where world knowledge biases them toward labeling plausible conclusions as valid. Subtask 1 focuses on English validity checking, while Subtask 2 adds noise through irrelevant premise identification. Subtasks 3 and 4 extend these challenges to a multilingual setting (11 languages), further combining validity classification with premise retrieval.

To address this, we compare a neural baseline against two translation pipelines mapping text to first order logic (FOL) or Aristotelian Syllogistic Triples (ASTs). By offloading inference to symbolic solvers, we decouple formal logic from semantic bias. In Subtask 1, our best model reached 95.29% accuracy, ranking 19th among 45. For Subtasks 2, 3, and 4, it achieved 95.79% (9th/16), 88.02% (12th/15), and 77.08% (11th/15), respectively.¹

2 Related Work

Belief bias, the cognitive tendency to favor semantic plausibility over logical validity, is a well-documented phenomenon in humans (Evans et al.,

1983) that persists in both early and contemporary large language models (Ando et al., 2023; Lampinen et al., 2024; Kim et al., 2025). Mechanistic analyses of models like Llama 3 and Qwen reveal that internal reasoning circuits are often ‘contaminated’ by attention heads dedicated to commonsense knowledge (Kim et al., 2025). To resolve this, our Translation-Based Inference via First-Order Logic (TBI-FOL) explicitly bypasses these biased circuits. By offloading logical deduction to the Otter theorem prover, we ensure that validity checks remain strictly formal and entirely independent of semantic content.

Recent research has explored various methods for modeling logical validity and syllogistic inference in natural language. Poddar et al. (2025) evaluated 14 different models and found that logical reasoning is not a uniform emergent property, with models often struggling with quantifier interpretation. Furthermore, Hua et al. (2025) observed that reasoning capabilities are often tied to specific linguistic domains rather than generalizable rules. To mitigate these effects, Valentino et al. (2025) proposed fine-grained activation steering to decouple content from logic. While steering shows promise in internal model adjustments, Zong and Lin (2024) argue that the vast variety of syllogistic moods still poses a significant challenge for purely neural architectures. These observations are consistent with our finding that compact models like SmolLM reach peak performance only when utilized as specialized structural parsers within a symbolic framework.

To achieve full disentanglement, neuro-symbolic pipelines have emerged as a robust alternative to purely neural methods. Xu et al. (2024) introduced Symbolic Chain-of-Thought (SymbCoT), proving that translating context into symbolic formats significantly reduces hallucinations. This paradigm is supported by Pan et al. (2023) and the need for faithful reasoning grounded in formal rules (Xu

¹Our code is available at <https://github.com/WiebkePetersen/HHU-SyLo-SemEval2026-Task11>.

et al., 2024). Our contribution builds on these insights by introducing a novel integration of LLM-based translation with established symbolic solvers. Unlike previous work, our pipeline utilizes Otter and Prolog to provide a definitive defense against content bias. By strictly separating linguistic translation from formal inference, our system effectively neutralizes the semantic interference identified in recent literature, providing a scalable solution for multilingual syllogistic reasoning.

3 System Overview

Our system implements a neuro-symbolic pipeline designed to decouple linguistic surface forms from formal logical reasoning. To address the challenge of content bias in syllogistic reasoning, we developed three distinct architectures for assessing argument validity (see 1).

Direct Neural Inference (DNI): A baseline approach where a Large Language Model (LLM) predicts binary validity directly from the natural language prompt without intermediate symbolic steps.

Translation-Based Inference via First-Order Logic (TBI-FOL): In this pipeline, natural language syllogisms are converted into First-Order Logic (FOL) formulas using two distinct translation paradigms: (1) *Neural Translation (Neur-TBI-FOL)*, where an LLM is prompted to generate FOL syntax, and (2) *Rule-based Translation (Rule-TBI-FOL)*, where a deterministic mapping is performed via a specialized Python-based regex engine to extract logical structures from the text.

The actual inference is then performed by the automated theorem prover *Otter*.² A syllogism is valid if Otter finds a contradiction between premises and the negated conclusion.

For Subtasks 2 and 4, we perform proof analysis by mapping the clause indices used in Otter’s refutation proof back to the original premises to identify the minimal set of relevant premises.

Translation-Based Inference via Syllogistic Triples (TBI-SYL): In this approach, the LLM maps the text to Aristotelian triples (q, S, P) , where $q \in \{a, e, i, o\}$ represents the categorical proposition type (e.g., “All S are P ” $\rightarrow$ ($'a'$, S , P)). Similar to TBI-FOL, this uses neural generation or rule-based extraction. Formal validity is

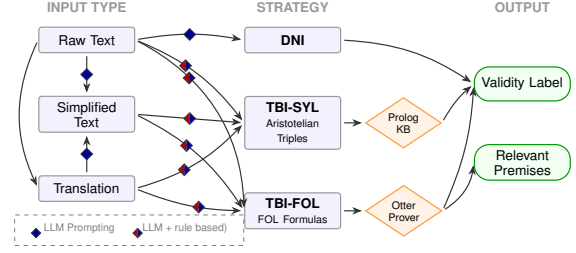


Figure 1: Overview of the modular neuro-symbolic pipeline. Arrows indicate the transition mechanism: purely LLM-based prompting (Blue Diamond) or two alternative transitions via either LLM or rule-based processing (Red and Blue Diamond).

text	It is obvious that not all dogs lay eggs.
simplified	some Dog is not LayEgg
TBI-FOL	exists x (Dog(x) & -LayEgg(x))
TBI-SYL	(o, Dog, LayEgg)

Figure 2: Illustrative example of inputs and outputs generated at each translation step.

then determined by a Prolog-based Knowledge Base encoding the 256 classical syllogistic moods.

The complete modular architecture and the various experimental paths, i.e. from input selection to final inference, are visualized in Figure 1. This design allows us to compare a wide spectrum of reasoning strategies anchored by two fundamental baselines: (I) A **purely neural path (DNI)**, where the model performs the entire task end-to-end, relying solely on its internal representations; and (II) a **purely symbolic path**, where our deterministic rule-based translation engine is applied directly to the raw text, bypassing LLM-mediated reasoning entirely. Figure 2 provides an illustrative example of the concrete outputs generated at each translation step.

The core motivation for employing external engines like *Otter* and the *Prolog KB* was the goal of ensuring that the final validity decision is strictly a function of the extracted logical structure. By offloading the inference to these formal solvers, we aimed to render the reasoning process immune to the semantic plausibility of the premises, thereby systematically eliminating the impact of content bias that often affects direct LLM predictions.

To further mitigate the impact of complex natural language, we implemented an optional *Linguistic Simplification* step. Here, an LLM simplifies raw text into standardized propositions with normalized predicates (e.g., “*Triangles are shapes with three sides*” $\rightarrow$ all Triangle is

²<https://www.cs.unm.edu/~mccune/otter/>

Shape_With_Three_Sides). The rationale behind this was to decouple the identification of conceptual predicates from the construction of a clean logical formula, thereby reducing the cognitive load on the subsequent translator.

To address the multilingual challenges (**Subtasks 3 and 4**), we compared two distinct strategies: (I) a direct application of the pipelines developed for Subtasks 1 and 2 to the native multilingual inputs, and (II) a two-stage approach where the syllogisms were first translated into English using Google Translator via `pypi deep-translator`³ before applying our reasoning pipelines.

4 Experimental Setup

To evaluate the robustness of our neuro-symbolic pipeline against content bias, we conducted extensive experiments across different tasks, model sizes, and prompting configurations.

Prompting Strategies The neuro-symbolic pipeline (*TBI-FOL* and *TBI-SYL*) faces two primary challenges: ensuring *terminological consistency* (mapping semantically equivalent concepts to identical predicates) and adhering to the *restrictive syntax* of the Otter prover and Prolog KB.

To address these, we developed specialized prompt categories for logic translation, syllogistic mapping, and linguistic simplification. These range from minimalistic, example-driven prompts to complex instructions with multilingual trigger word mappings for Subtasks 3 and 4. A baseline *Direct Neural Inference* (DNI) prompt was also used for direct one-step binary validity judgments.

The prompts are supported by curated few-shot example sets, which are balanced across belief-bias conditions (validity vs. plausibility) to guide the models through the logical transformations. Detailed descriptions of all prompt variants and the composition of the few-shot sets are provided in Appendix A.1.

4.1 Models and Parameters

To investigate scaling effects and reasoning capabilities within the category of compact Large Language Models (LLMs), we evaluated our pipeline across a diverse selection of models ranging from 135M to 4B parameters. The selection includes the

SmolLM series, chosen as a fully open model family (providing transparency regarding training data mixtures and recipes) while being specifically optimized for reasoning tasks at a small scale, the **Qwen** family, which was selected for its established strength in formal logic and its native multilingual capabilities (crucial for Subtasks 3 and 4). Additionally, **Llama** and **Gemma** models serve as industry-standard benchmarks for instruction-following and state-of-the-art reasoning in the 2B to 4B parameter range.

To ensure deterministic and reproducible outputs, we primarily employed greedy decoding (temperature 0.0), with minor model-specific adjustments to ensure generation stability where required. Detailed model identifiers and the complete hyperparameter configuration are provided in Appendix A.2.

4.2 Data Processing and Symbolic Integration

To bridge the gap between natural language and formal logic, we implemented a processing pipeline that handles output extraction, deterministic rule-based translation, and the interface to symbolic solvers. Detailed extraction steps and prover configurations are provided in Appendix A.3.

Normalization and Rule-based Translation LLM output undergoes aggressive artifact stripping and term normalization (e.g. article removal and singularization), to ensure terminological consistency across premises. For the non-neural path, a deterministic engine maps propositions to FOL via copula identification and hierarchical regex patterns. By offloading the formalization, we guarantee valid FOL syntax, effectively mitigating the risk of malformed outputs often produced by LLMs. This allows the system to remain robust whether the LLM is used only for linguistic simplification or is bypassed entirely.

Inference and Existential Import To align FOL with the existential import of classical syllogisms, the pipeline automatically injects existence conditions ($\exists xP(x)$) for all detected predicates. Formulas are further standardized to meet the strict syntactic requirements of the *Otter* theorem prover, including alpha conversion, negation normalization and quantifier scoping. For Subtasks 2 and 4, we utilize the resolution proof history from the prover to identify the minimal sufficient set of premises.

³`pypi deep-translator`: <https://pypi.org/project/deep-translator/>

Syllogistic Path (TBI-SYL) The triple-based pipeline maps propositions to (q, S, P) structures, which are validated against a Prolog knowledge base containing the 24 valid Aristotelian moods. Unlike the FOL path, this solver accounts for existential import through its internal encoding.

4.3 Evaluation Metrics

We evaluate our systems using the official metrics defined for the SemEval-2026 Shared Task 11 (Valentino et al.). The evaluation focuses on three dimensions: (1) **Accuracy (Acc)** for validity prediction in Subtask 1, (2) the **Macro-Averaged F1-Score** ($F1_{prem}$) for premise identification in Subtask 2, and (3) the **Total Content Effect (TCE)**, which quantifies belief bias by measuring performance disparities across plausible and implausible subgroups. To determine the final ranking, a **Combined Score** is calculated by applying a logarithmic penalty to the average of Subtask 1 and 2 performance, weighted by the observed TCE.

5 Results and Discussion

This section evaluates the performance of our modular architecture across all shared task challenges on the subtasks’ test sets provided by the shared task. Due to its role as the logical foundation for all subsequent reasoning steps, we first provide an extensive analysis of Subtask 1, followed by an evaluation of the system’s performance in premise selection and multilingual settings. For a comprehensive overview of all tested experimental configurations see Appendix B.

5.1 Subtask 1: Logical Validity

The primary objective of Subtask 1 is to establish a robust backbone for validity prediction while minimizing belief bias. We conducted 31 experimental configurations comparing architectures, model families, scaling effects, and prompting strategies. Our analysis identifies **Eng2FOL5** as the superior configuration, utilizing the *SmolLM3-3B* model with an *Ultrashort Prompt* (320 tokens) and eight varied examples. It serves as the benchmark for all subsequent comparisons.

Architecture Comparison and Baselines We compare purely neural (DNI) and symbolic (Rule) baselines against various hybrid *Translation-Based Inference* (TBI) approaches. These hybrids differ in their translation mechanism (LLM vs. Rule), the input format (Raw vs. Simplified), and the logical

target (FOL vs. SYL). All experiments reported in Table 1 were conducted using the *SmolLM3-3B* model as the primary engine for both direct inference and translation tasks.

Architecture	Acc	TCE	Score
<i>Baselines</i>			
DNI (Direct Neural Inference)	56.54	24.51	13.34
Rule (Rule-TBI-FOL/Syl)	49.74	–	–
<i>Hybrid TBI</i>			
<i>from raw syllogism</i>			
LLM-TBI-FOL (Eng2FOL)	91.62	4.17	34.68
LLM-TBI-FOL (Eng2FOL5)	95.29	2.15	44.37
LLM-TBI-Syl (Eng2SyllogismProlog)	72.25	18.66	18.16
<i>from simplified syllogism</i>			
Simp(1)-Rule-TBI-FOL	86.39	14.58	23.06
Simp(1)-Rule-TBI-Syl	85.34	18.75	21.43
Simp-LLM-TBI-FOL (SimpEng2FOL1)	85.86	12.50	23.83
Simp-LLM-TBI-Syl (SimpEng2SyllogismProlog)	76.96	14.49	20.58

Table 1: Architecture Comparison: Performance of Neural, Symbolic, and Hybrid Paradigms using *SmolLM3-3B*.

Both baseline approaches perform poorly. The symbolic baseline (*Rule*) achieves an accuracy of only $\approx 50\%$, failing to find a single valid proof. This stems from the fact that the rule-based grammar was tailored for simplified English; when faced with raw natural language, the extraction of syntactically well-formed formulas fails in 121 out of 191 cases. Similarly, the *Direct Neural Inference* (DNI) baseline, despite a strict prompt emphasizing logical validity over plausibility, achieves only 56% accuracy, with 25 cases being excluded as errors due to non-compliant output formats.

In contrast, the efficacy of the rule-based system increases dramatically when integrated into a hybrid pipeline. When applied to syllogisms previously simplified by *SmolLM3-3B*, the rule-based translation (*Simp-Rule-TBI*) reaches accuracies of $\approx 86\%$ for both FOL and SYL (Aristotelian Triples) targets.

When delegating the translation entirely to the LLM, a significant performance gap emerges between logical targets. *LLM-TBI-FOL* consistently outperforms *LLM-TBI-Syl*. This advantage likely results from the higher prevalence of FOL-like structures in the LLM’s training data. Furthermore, FOL offers higher flexibility, as multiple logically equivalent translations (e.g., varying term orders or conjunctions) can be correctly resolved by the FOL

theorem prover, whereas the triple-based SYL format requires an exact match of subject, predicate, and mood. Due to this robustness and the superior traceability of premises in FOL proofs (crucial for Subtask 2) we focused our subsequent optimization on FOL-based translation.

Against our intuition, the use of a two-stage LLM pipeline (1st simplification, 2nd translation into FOL) did not yield improvements. Comparing *Eng2FOL* (raw) and *SimpEng2FolI* (simplified) using identical prompts, we observe an accuracy drop and a sharp increase in the *Total Content Effect* (TCE) from 4.17 to 12.5. This suggests that simplification introduces more noise than it removes. Crucially, the 3B-parameter models already produced surprisingly clean FOL syntax from raw text, making an intermediate stage redundant. Consequently, our best system, *Eng2FOL5*, builds directly on the single-stage-translation *Eng2FOL* approach using optimized prompts and few-shot examples.

Impact of Model Selection and Prompting Strategies Our evaluation across various LLM families and prompting configurations reveals that while smaller models like *SmolLM3-3B* and *Qwen3-4B* can achieve high accuracy ($> 95\%$), the results remain notably unstable. Performance metrics fluctuate significantly depending on the specific few-shot example selection and the alignment between prompt length and model architecture. We observe that no single prompting strategy yields a uniform advantage. Furthermore, a model’s peak performance on one subtask or dataset does not consistently translate to others, highlighting a sensitivity to input distributions that complicates the search for a universal "winning" configuration. A comprehensive breakdown of model scaling effects, prompt variations, and full performance tables is provided in Appendix B.

5.2 Subtasks 2–4: Premise Identification and Multilingual Robustness

For **Subtask 2**, premise identification is natively handled by our TBI approach, as the FOL theorem prover (Otter) generates a resolution proof that explicitly lists all utilized premises. The primary challenge lies in the consistent handling of existence clauses introduced during preprocessing.

The best results for Subtask 2 were achieved by *Eng2FOL5Qwen4B*, using the *Qwen3-4B-Instruct* model with the same few-shot configuration as our winning Subtask 1 setup (Acc: 95.79%, F_1 -

Premises: 92.37%, TCE: 3.26, Score: 38.43). This performance underscores the high sensitivity of smaller LLMs to specific input distributions: while *SmolLM3-3B* dominated Subtask 1, it only reached 86.98% accuracy (F_1 : 78.38%) on the Subtask 2 test set using the identical configuration (*Eng2FOL5*).

For the multilingual challenges (**Subtasks 3 and 4**), we compared two strategies: (I) automated translation into English prior to inference, and (II) direct processing of multilingual inputs.

In **Subtask 3**, approach (I) proved slightly superior with the *QwenEng2FOLmultilingualNewShort* configuration (Acc: 88.02%, TCE: 6.38, Score: 29.36). However, the direct multilingual approach (II) using the same *Qwen3-4B-Instruct* model (*QwenEng2FOL6*) performed nearly as well, achieving a higher raw accuracy of 90.63%, albeit with a higher TCE (7.33).

In **Subtask 4**, the English-translation strategy (I) combined with the winning configuration from Subtask 1 (*Eng2FOL5*) yielded the best results (Acc: 77.08%, F_1 -Premises: 64.15%, TCE: 6.07, Score: 23.89). Direct multilingual processing suffered significantly in this subtask; the corresponding *Qwen* configuration showed a markedly higher content effect (TCE: 14.71).

6 Conclusion

We presented a modular hybrid neuro-symbolic architecture for reasoning disentanglement. By contrasting pure neural and pure symbolic approaches with hybrid ones, we demonstrated that relatively small specialized LLMs like *SmolLM3-3B* can achieve high logical fidelity when the LLM’s linguistic flexibility is anchored by the formal rigor of a symbolic solver.

Future work should address identified limitations, particularly the alignment between prompt instructions and few-shot examples. Refined rule-based post-processing and more rigorous prompt optimization could further stabilize the translation layer, ensuring that hybrid neuro-symbolic pipelines remain robust across diverse linguistic and logical contexts.

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A Experimental setup details

A.1 Detailed Prompting Configurations

This section provides the technical specifications of the prompts and few-shot examples used in our experiments.

Prompts All full prompt templates are available in our repository: <https://github.com/WiebkePetersen/HHU-SyLo-SemEval2026-Task11/tree/main/prompts>.

- **Logic Translation (TBI-FOL):** We compared three prompt variants: Eng2FOL (focusing on explicit syntactic rules, 1721 chars), Eng2FOLimproved (emphasizing semantic decoupling, abstract treatment of symbols, and consistent PascalCase naming, 1634 chars), and Eng2FOLultrashort (a minimalistic, example-driven approach, 320 chars).

Multilingual TBI-FOL: Two dedicated prompts, MultiLingFOL (3033 chars) and MultiLingEng2FOLultrashort (502 chars), handled the multilingual scenarios (Subtasks 3 and 4). MultiLingFOL includes a step-by-step reasoning instruction (in English) and a multilingual trigger word mapping for 11 languages to ensure logical fidelity during the *Native-test* condition.

- **Syllogistic Mapping (TBI-SYL):** The SyllTrip prompt (1254 chars) maps natural language to Aristotelian triples (q, S, P), enforcing naming consistency and mapping quantifiers to categorical proposition types (a, e, i, o).
- **Linguistic Simplification:** Two prompts, LogSimp (1572 chars) and LogSimpShortPrompt (1016 chars), were employed to evaluate the impact of text normalization. These prompts instruct the LLM to extract a standardized "Simplified Logic English" (e.g., all [Subject] is [Predicate], see Figure 2), enforcing singular forms capitalization, atomization

for multi-word terms, and the elimination of negative atoms (e.g., converting "is not a fish" to a "no" quantifier and not to "not_fish").

Few-Shot Example Sets Table 2 details the composition of the curated example sets used to guide the models. It provides details regarding the quantity, average length, and the distribution of validity and plausibility across the example sets.

Example Set ID	#	IV	PV	II	PI	$\emptyset$
LogSimpEx	5	1	4	0	0	3.4
SyllTripEx	4	1	3	0	0	3.8
SimpSyllTripEx	4	1	3	0	0	3.8
Eng2FOLex	3	0	0	1	2	3.0
Eng2FOLexImproved	5	0	0	2	3	3.2
Eng2FOLexImprovedX	8	0	3	2	3	3.1
Eng2FOLexImproved2	8	2	2	2	2	3.6
Eng2FOLexVaried	8	0	5	1	2	3.0
Eng2FOLexLong	4	0	2	2	0	7.5
MultiLingExImproved	22	2	9	1	10	3.0
MultiLingExShort	6	1	2	0	3	3.0

Table 2: Composition of few-shot example sets. # denotes the total number of examples; $\emptyset$ represents the average number of logical statements (premises and conclusion) per syllogism. Categories reflect belief-bias conditions: Implausible/Valid (IV), Plausible/Valid (PV), Implausible/Invalid (II), and Plausible/Invalid (PI).

A.2 Model Specifications and Hyperparameters

The following specific LLM checkpoints were utilized in our experiments:

- **SmolLM-Series:** SmolLM2-135M-Instruct, SmolLM2-1.7B-Instruct, and SmolLM3-3B.
- **Qwen-Series:** Qwen3-0.6B, Qwen3-1.7B, and Qwen3-4B-Instruct-2507.
- **Gemma-Series:** google/gemma-2-2b-it and google/gemma-3-4b-it.
- **Llama-Series:** meta-llama/Llama-3.2-3B-Instruct.

To ensure reproducibility, we set the temperature to 0.0 for the majority of models. For the *Qwen* models, a specific configuration was used to ensure generation stability: `do_sample=True`, `temperature=0.1`, `top_p=0.9`, and a `repetition_penalty` of 1.1. Inference was performed with batch sizes between 8 and 16, depending on the respective model size.⁴

⁴A comprehensive list of all hyperparameters and model-specific settings is provided in the `experiments.yaml` configuration file in our repository.

A.3 Implementation Details: Processing and Translation

Extraction and Normalization

Since LLMs often embed formal outputs within conversational text, we employ a multi-step extraction layer involving recursive JSON and literal parsing to isolate logical lists. Artifact stripping (e.g., removing trailing explanations) is applied to ensure clean inputs for the solvers. During normalization, phrases like "*the birds*" and "*a bird*" are both mapped to the unified predicate `Bird`. All terms are capitalized, singularized and atomized.

Rule-based Translation Engine

The deterministic engine first attempts a heuristic split based on copula identification ("*is*", "*are not*"). It specifically handles complex linguistic cases like double negations (e.g., "*no X is not Y*"). If a direct split fails, a hierarchical regex safety net identifies the categorical proposition type.

Symbolic Standardization

Before inference, formulas undergo refinement for *Otter* compatibility:

- **Alpha-conversion:** Ensures unique variable naming across the proof.
- **Syntactic Cleaning:** Parentheses balancing and negation standardization.
- **Syllogistic Existential Closure Assumption:** Injection of existence conditions ($\exists x P(x)$) for all detected predicates.
- **Negated Conclusion:** The conclusion is identified and provided to the prover in its negated form to facilitate a proof by contradiction.

For successful refutations, the *Otter*'s proof history is parsed to map used clauses back to their original premises.

B Detailed Results: Models and Prompting

Table 5 shows the results of all our experiments run on the test data of Subtask 1.

Impact of Model Selection and Scaling on Performance

Model Scaling and Family Performance

To evaluate the impact of model capacity under controlled conditions, we compared various LLMs using the prompting configuration (*Ultrashort Prompt*, 8 examples) from our winning experiment (Eng2FOL5). As shown in Table 3, *SmolLM3-3B* achieves the highest combined score, marginally outperforming the *Qwen* models due to its superior bias mitigation (lowest TCE). It also significantly outperforms both *Llama* and *Gemma* in this task. While we observe a general scaling trend where larger models yield better results, the *Qwen3-1.7B* model represents a notable exception, demonstrating remarkably high accuracy for its size.

Model (Experiment)	Comb	Acc	TCE
SmolLM3-3B (Eng2FOL5)	44.37	95.29	2.15
Qwen3-4B-Inst. (Eng2FOL5Qwen4B)	36.23	96.34	4.26
Qwen3-1.7B (Eng2FOL5Qwen1.7B)	36.03	95.81	4.26
Llama-3.2-3B-Inst. (Eng2FOL3)	24.89	85.34	10.35
Gemma-2-2B-it (Eng2FOL5Gemma2B)	20.08	73.82	13.54
SmolLM2-1.7B-Inst. (Eng2FOL5small)	17.61	65.97	14.58
Qwen3-0.6B (Eng2FOL5Qwen0.6B)	17.53	65.45	14.38
SmolLM2-135M-Inst. (Eng2FOL5tiny)	11.70	50.26	26.04

Table 3: Impact of model size and family on TBI-FOL performance using a fixed configuration (Eng2FOLUltrashort prompt, Eng2FOLexImprovedX examples; details in Appendix A.1).

While Table 3 compares LLMs under identical conditions, Table 4 reports the peak performance for each LLM across all tested prompt configurations. While most architectures benefit from the *Ultrashort* setup, individual tuning can further reduce the TCE for specific families like *Qwen* (e.g., *QwenEng2FOL1*).

Model (Best Experiment)	Comb	Acc	TCE
SmolLM3-3B (Eng2FOL5)	44.37	95.29	2.15
Qwen3-4B-Instruct-2507 (QwenEng2FOL1)	38.44	93.72	3.21
Qwen3-1.7B (Eng2FOL5Qwen1.7B)	36.03	95.81	4.26
Llama-3.2-3B-Instruct (Eng2FOL3)	24.89	85.34	10.35
gemma-2-2b-it (Eng2FOL5Gemma2B)	20.08	73.82	13.54
SmolLM2-1.7B-Instruct (Eng2FOL5small)	17.61	65.97	14.58
Qwen3-0.6B (Eng2FOL5Qwen0.6B)	17.53	65.45	14.38
SmolLM2-135M-Instruct (Eng2FOL5tiny)	11.70	50.26	26.04
gemma-3-4b-it (Eng2FOL11)	10.41	43.46	22.92

Table 4: Peak performance per model family across all experimental configurations. Each entry represents the best-performing individual setup for that specific architecture.

Prompting Strategies and Few-Shot Effects

The transition between prompt strategies yields no uniform advantage across all architectures.

For *SmolLM3-3B*, we observe a slight preference for shorter prompts (see Appendix A.1 for prompt details): comparing Eng2FOLUltrashort (Eng2FOL5) to the longer Eng2FOL (Eng2FOL7), accuracy marginally increases from 95.29% to 96.34%, yet this is offset by the *Total Content Effect* (TCE) doubling from 2.15 to 4.26 (see Table 5 for full details). Conversely, *Qwen* models exhibit an opposite tendency, showing improved robustness with longer, more descriptive prompts (e.g., QwenEng2FOL4 vs. QwenEng2FOL2).

Variations in the few-shot example sets similarly provide an ambiguous picture (see Table 2 for overview of used few-shot example sets). Analysis of experiments Eng2FOL5, 6, 8, and 9—which differ solely in example selection—indicates a subtle trend where a higher quantity of examples improves performance. However, explicitly balancing the plausibility-validity distribution within the examples did not yield the expected gains in bias mitigation.

ExpID	FP	FN	Errors	Acc	TCE	Score	Acc _{PV}	Acc _{PI}	Acc _{II}	Model	Prompt	PromptLen	Input	ExID	#Ex	ExLen
Eng2FOL5	4	5	0	95.29	2.15	44.37	93.75	95.83	93.62	97.92	Eng2FOLUltrashort	320	syll	Eng2FOLexImprovedX	8	3.12
QwenEng2FOL3	5	7	12	93.72	2.08	44.08	91.67	93.75	93.62	95.83	Eng2FOLUltrashort	320	syll	Eng2FOLex	3	3.00
QwenEng2FOL4	3	2	1	97.38	3.19	40.02	95.83	100.00	93.62	100.00	Eng2FOLImproved	1634	syll	Eng2FOLexImproved	5	3.20
Eng2FOL7	4	2	2	96.86	3.19	39.81	95.83	100.00	93.62	97.92	Eng2FOL	1721	syll	Eng2FOLexImprovedX	8	3.12
Eng2FOL5Qwen4B	3	3	1	96.86	3.19	39.81	93.75	100.00	93.62	100.00	Eng2FOLUltrashort	320	syll	Eng2FOLexImprovedX	8	3.12
Eng2FOL5Qwen1.7B	3	4	1	96.34	3.19	39.59	95.83	95.83	93.62	100.00	Eng2FOLUltrashort	320	syll	Eng2FOLexImprovedX	8	3.12
Eng2FOL6	5	5	1	94.76	3.21	38.86	91.67	97.92	91.49	97.92	Eng2FOLUltrashort	320	syll	Eng2FOLexImproved	5	3.20
QwenEng2FOL5	3	5	1	95.81	4.17	36.26	91.67	97.92	93.62	100.00	Eng2FOL	1721	syll	Eng2FOLexImproved	5	3.20
Eng2FOL10Qwen4B	3	5	0	95.81	4.17	36.26	91.67	97.92	93.62	100.00	Eng2FOLImproved	1634	syll	Eng2FOLexVaried	8	3.00
QwenEng2FOL2	3	6	5	95.29	4.17	36.06	91.67	95.83	93.62	100.00	Eng2FOLUltrashort	320	syll	Eng2FOLexImproved	5	3.20
QwenEng2FOL1	3	7	3	94.76	4.17	35.87	91.67	93.75	93.62	100.00	Eng2FOLUltrashort	320	syll	Eng2FOLexImproved2	8	3.62
Eng2FOL	5	11	2	91.62	4.17	34.68	89.58	87.50	93.62	95.83	Eng2FOL	1721	syll	Eng2FOLex	3	3.00
QwenEng2FOL6	3	5	0	95.81	5.21	33.90	89.58	100.00	93.62	100.00	Eng2FOLImproved	1634	syll	Eng2FOLexImprovedX	8	3.12
Eng2FOL8	5	12	1	91.10	5.21	32.24	87.50	87.50	91.49	97.92	Eng2FOLUltrashort	320	syll	Eng2FOLexImproved2	8	3.62
Eng2FOL4	5	11	1	91.62	6.25	30.74	85.42	91.67	91.49	97.92	Eng2FOLImproved	1634	syll	Eng2FOLexImproved	5	3.20
Eng2FOL9Qwen4B	3	8	12	94.24	7.29	30.25	85.42	97.92	93.62	100.00	Eng2FOLUltrashort	320	syll	Eng2FOLexVaried	8	3.00
Eng2FOL3	9	19	2	85.34	10.35	24.89	72.92	87.50	93.62	87.50	Eng2FOLUltrashort	320	syll	Eng2FOLexImprovedX	8	3.12
Eng2FOL10	4	20	2	87.43	12.50	24.27	72.92	85.42	93.62	87.92	Eng2FOLImproved	1634	syll	Eng2FOLexVaried	8	3.00
Eng2FOL9	5	21	3	86.39	12.50	23.98	70.83	85.42	93.62	95.83	Eng2FOLUltrashort	320	syll	Eng2FOLexVaried	8	3.00
SimpEng2FOL1	5	22	2	85.86	12.50	23.83	70.83	83.33	93.62	95.83	Eng2FOL	1721	Simp1	Eng2FOLex	3	3.00
Simp1-Rule-TBI-FOL	2	24	3	86.39	14.58	23.06	70.83	79.17	95.74	100.00	LogSimp	1572	syll	LogSimpEx	5	3.40
Simp2-Rule-TBI-FOL	6	24	3	84.29	13.54	22.92	70.83	79.17	89.36	97.92	LogSimpShortPrompt	1016	syll	LogSimpEx	5	3.40
Eng2FOL2	14	31	5	76.44	9.38	22.89	68.75	66.67	85.11	85.42	Eng2FOLImproved	1634	syll	Eng2FOLexImproved	5	3.20
SimpEng2FOL2	6	25	1	83.77	14.58	22.36	68.75	79.17	89.36	97.92	Eng2FOL	1721	Simp2	Eng2FOLex	3	3.00
Simp1-Rule-TBI-Syl	0	28	0	85.34	18.75	21.43	62.50	79.17	100.00	100.00	LogSimp	1572	syll	LogSimpEx	5	3.40
SimpEng2SyllogismProlog	9	35	0	76.96	14.49	20.58	64.58	62.50	91.49	89.58	SyllTrip	1254	Simp1	SimpSyllTripEx	4	3.75
Simp2-Rule-TBI-Syl	4	29	0	82.72	19.79	20.50	58.33	81.25	93.62	97.92	LogSimpShortPrompt	1016	syll	LogSimpEx	5	3.40
Eng2FOL5Gemma2B	14	33	6	75.39	16.67	19.47	60.42	70.83	76.60	93.75	Eng2FOLUltrashort	320	syll	Eng2FOLexImprovedX	8	3.12
Eng2SyllogismProlog	13	40	0	72.25	18.66	18.16	54.17	62.50	91.49	81.25	SyllTrip	1254	syll	SyllTripEx	4	3.75
Eng2FOL5small	23	41	10	66.49	14.58	17.75	62.50	52.08	70.21	81.25	Eng2FOLUltrashort	320	syll	Eng2FOLexImprovedX	8	3.12
Eng2FOL5Qwen0.6B	17	43	19	68.59	17.58	17.49	52.08	58.33	87.23	77.08	Eng2FOLUltrashort	320	syll	Eng2FOLexImprovedX	8	3.12
Eng2FOL11	0	69	100	63.87	37.50	13.73	25.00	31.25	100.00	100.00	Eng2FOLUltrashort	320	syll	Eng2FOLexImprovedX	8	3.12
DNI	45	38	25	56.54	24.51	13.34	79.17	41.67	46.81	58.33	-	-	-	-	-	-
Eng2FOL5tiny	62	29	10	52.36	23.96	12.41	70.83	68.75	46.81	22.92	Eng2FOLUltrashort	320	syll	Eng2FOLexImprovedX	8	3.12
Rule-TBI-Syl	0	96	126	49.74	50.00	10.09	0.00	0.00	100.00	100.00	-	-	-	-	-	-
Rule-TBI-FOL	0	96	121	49.74	50.00	10.09	0.00	0.00	100.00	100.00	-	-	-	-	-	-

Table 5: Detailed Performance Metrics per Model and Configuration on test dataset for Subtask 1.